

Reflection on Abstract Art through Kandinsky's Teaching

Reflection on abstract art through reconstruction of Kandinsky's teaching at the Bauhaus

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ABSTRACT

Wassily Kandinsky is considered as one of the pioneers of abstract painting in Europe. In this paper, our goal is to develop a user interface that models Kandinsky's analytical drawing process within a computational context to advance research in artificial intelligence and the arts. A user first chooses a photograph from the Object-Net3D database, objects of which are aligned with the 3D shapes. Using the 3D shape information, the software extracts 2D projection contours. Then, it performs corner detection and finds structural networks. Lastly, the software transforms the photograph into an abstract image based on a student example.

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Fine arts.

KEYWORDS

Kandinsky, Abstract art, Analytical drawing, Image stylization

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1 INTRODUCTION

In recent years, just as artists are learning programming to make art, engineers have found many ways to apply Artificial Intelligence (AI) algorithms to generate art. However, as there's a world of difference between art and science/engineering in the methods of approaching research, a careful investigation on the artist's intention or process of making art is needed when art and technology are combined [Maycock et al., 2009].

Neural style transfer (NST), a class of artificial systems based on a Deep Neural Network that creates artistic im-ages of high perceptual quality, was first introduced in Gatys et al.'s work [Gatys et al., 2015]. There have been significant efforts made to synthesize

stylized images by using neural networks since 2015 when this seminal work was released.

Although these algorithms successfully produced stylized images with the appearance of a given artwork, defining style related to texture does not guarantee the quality of artwork [Alexandru et al., 2022]. We created software that models Kandinsky's analytical drawing process, which was used for formulating a composition in many of his abstract paintings and investigated the fundamental limitation of NST in this paper.

2 THEORY OF ANALYTICAL DRAWING (SUMMARY OF ANALYTICAL DRAWING IN ANALYTICAL DRAWING [Poling, 1986])

Analytical drawing was part of Kandinsky's preliminary course taught at the Bauhaus. As Kandinsky explained in his article "Analytical Drawing", "the teaching of drawing at the Bauhaus is an education in looking, precise observation, and the precise representation not of the external appearance of an object, but of constructive elements and of their logical construction."

Kandinsky mentioned that this training would develop students' ability to perceive the abstract and the essential form. He used still life as an artistic medium in analytical drawing because of its transitional role in the evolution of abstraction. The analytical drawings provide the transitional link between still life and the abstract. They contributed to the formation of a new sensitivity, training the student to see the relationships among forms.

As outlined in Kandinsky's 1928 article, analytical drawing was a process in three stages: simplification, analysis, and transformation. See the appendix for the detailed process. Exercises made by Bella Ulmann-Broner in ref. [Poling, 1986] are used as a basis for analytical drawing research. Consecutive images in the book presented the simplification, network of tensions, principal tension (Fig. 98 in ref. [Poling, 1986]), and the transformation part (Fig. 124 in ref. [Poling, 1986]) of analytical drawing. In this paper, we focus our analysis on this student's style of work and proceed further discussion based on it.

3 IMPLEMENTATION

We have approached the implementation of analytical drawing as in the following. The software allows users to select images from the ObjectNet3D database, objects of which are aligned with the 3D shapes. Using the 3D shape information, the software then extracts 2D projection contour (simplification). Then, it executes corner detection and finds structural networks (analysis). For the transformation part, although results can come in many different forms for each person, the software shows an example of how Bella



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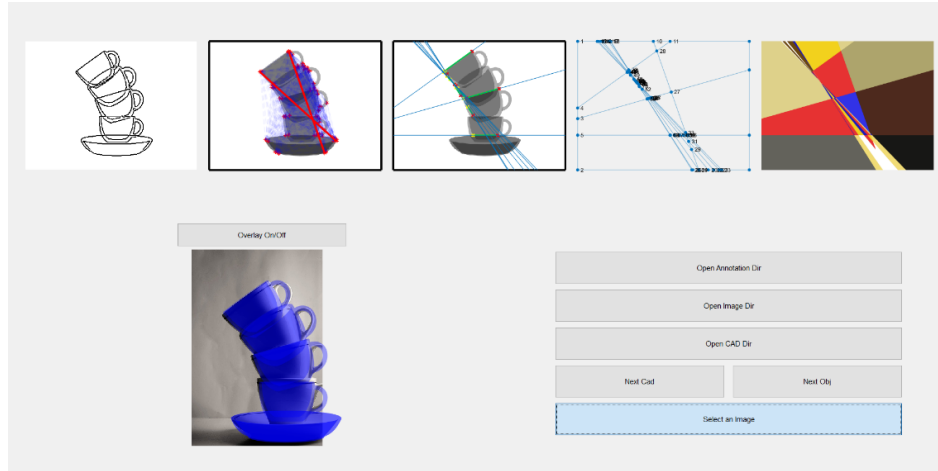


Figure 1: GUI for the program. (©Jungah Son. Photo: GUI for analytical drawing.)

Table 1: 100 object categories in the ObjectNet3D database

aeroplane	astray	backpack	basket	bed	bench	bicycle
blackboard	boat	bookshelf	bottle	bucket	bus	cabinet
calculator	camera	can	cap	car	cellphone	chair
clock	coffee maker	comb	computer	cup	desk lamp	diningtable
dishwasher	door	eraser	eyeglasses	fan	faucet	filing cabinet
fire extinguisher	fish tank	flashlight	fork	guitar	hair dryer	hammer
headphone	helmet	iron	jar	kettle	key	keyboard
knife	laptop	lighter	mailbox	microphone	microwave	motorbike
mouse	paintbrush	pan	pen	pencil	piano	pillow
plate	pot	printer	racket	refrigerator	remote control	rifle
road pole	satellite dish	scissors	screwdriver	shoe	shovel	sign
skate	skateboard	slipper	sofa	speaker	spoon	stapler
stove	suitcase	teapot	telephone	toaster	toilet	toothbrush
train	trash bin	trophy	tub	tvmonitor	vending machine	washing machine
watch	wheelchair					

Ullmann-Broner might have transformed the networks. Figure 1 shows a GUI prototype that we have created for showing the steps of analytical drawing.

4 THE SUBJECT MATTER OF ABSTRACT ART

Animating objects such as tables, chairs, and baskets is not a simple problem since these objects are mute and have a lack of motion [Lundmark, 2017]. However, artists find such “discreet, silent, almost insignificant objects” internally resonant as Kandinsky put it, and attribute great importance to this quality. Accordingly, still lifes such as simple furniture and building materials were used as the subjects in his class. To simulate the analytical drawing process, we tried to find a database which consists of images in which objects are aligned with the 3D shapes.

Such databases have been introduced in recent years due to the increased interest in 3D object recognition. To cover many object categories, we have chosen the Object-Net3D [Xiang et al., 2016],

which is a large-scale database that consists of 100 categories in Table 1. The database contains only rigid objects, which is also consistent with the fact that most of the objects used in analytical drawing were rigid.

4.1 Color

Kandinsky emphasized the psychic effect of colors in his book *Concerning the Spiritual in Art* [Kandinsky, 1977]. He introduced two divisions of color that occur to the mind at the outset: into warm and cold, and into light and dark.

The “cold colors” such as blue seem to recede, to move away from the spectator, while the “warm colors” such as yellow seem to come close to the spectator. As Kandinsky explained the effects of color using simple colors (e.g., yellow & blue, white & black), we found it easy to simulate colors in RGB space.

In addition, since the intersections of the lines drawn from the steps of analytical drawing are added as nodes to a graph, it was

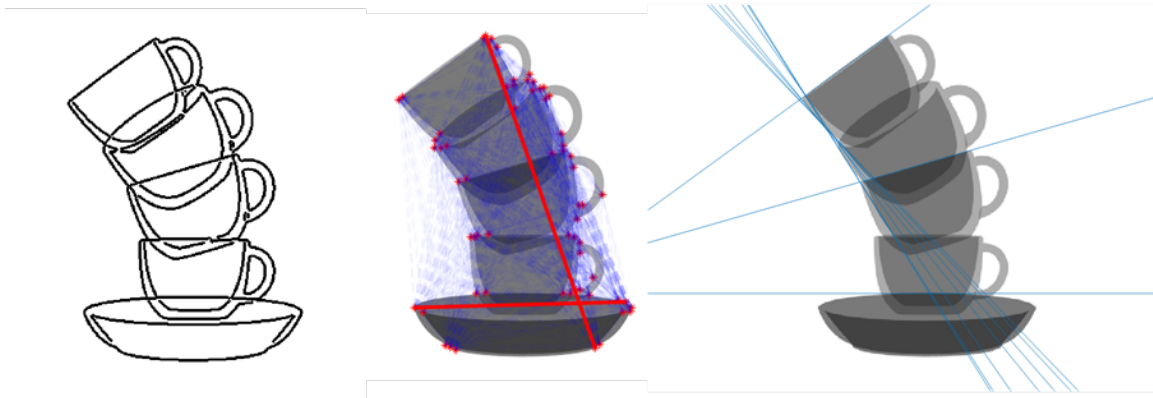


Figure 2: Edge detection (left), corner detection (middle), and Hough lines (right). (©Jungah Son.)

not hard to change the color of the regions to see different color treatments of the network.

4.2 Line

As well-represented in Ullmann-Broner's work, analytical drawing begins with flat outline drawings and progresses towards the linear analysis of the tensions and their relationships discovered in the structure. The outline or the contour of the objects illustrated in the simplification stage is implemented as edge detection (Figure 2, left).

While edge detection is obtainable directly from the image, developing the structural network (tensions) in the analysis stage required interpretation. In science/engineering, this problem can be approached by mapping suitable 3D models for the objects in the image and doing actual physics simulations to calculate tensions. However, in visual arts, tensions are referred to as psychological forces. [Arnheim, 1967] The term "movement" or "motion" had been used to describe visual dynamics, and Kandinsky replaced the almost universally accepted concept 'movement' with tension. [Arnheim, 1967]

As there is a difference between physical forces and psychological forces, here we focus on visual dynamics, i.e., constructive elements, rather than the external appearance of an object. After analyzing these dynamic properties, we used the following steps to find visual tensions in an image. First, corners in the image are detected. Then, among the lines connecting each corner, the longest tension line and a line that has opposite slope are defined as principal tensions. (Figure 2, middle) In addition, Hough transform, a feature extraction technique used to find imperfect instances of objects within a certain class of shapes by a voting procedure, is performed to detect straight lines in the image (Figure 2, right). These lines are considered as secondary tensions as can be seen in Ullmann-Broner's work. While constructing these hidden relationships among forms, we got to know the function of lines other than the outline.

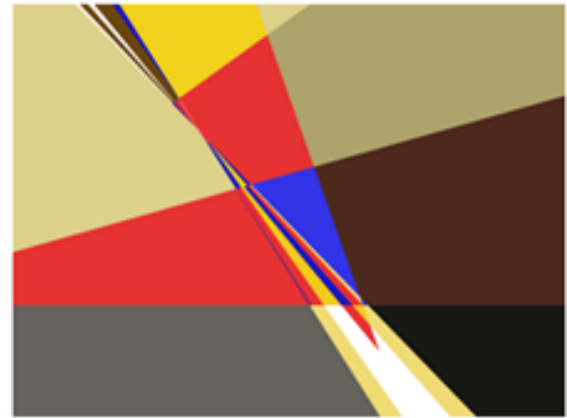


Figure 3: Transformation based on Ullmann-Broner's colored treatment of a network. (©Jungah Son.)

4.3 Composition and Harmony

Transformation, the third stage of analytical drawing, reveals what Kandinsky might well have suggested to his Bauhaus students for formulating a composition. One possibility of composition was an extremely flat image that fills the rectangular field, exploiting the diagonal in the initial "representation" and rendering the set of abstract relationships diagrammed in the "network" as shown in Ullmann-Broner's work.

In this case, harmony, expressive nature of the wholeness of artwork, was achieved by filling in the network with flat areas of color [Barasch, 2013]. Considering that the third stage allows most radical, freer abstract solutions, we have tried Kandinsky's way of finding harmony by balancing the contradictions, i.e., compositional use of contrasting pairs of colors such as yellow-blue balance. Figure 3 is an example composition produced by following Ullmann-Broner's variation.

5 CONCLUSION

In this paper, we implemented Kandinsky's analytical drawing approach into a user interface that shows the steps of the drawing



Figure 4: A content image (left), a style reference image (middle), and the output image (right). (From TensorFlow Tutorials. License CC BY 4.0 DEED)

process, and the analytical drawing approach allows geometrical simplifications and abstractions from the motif [Poling, 1986]. Although the resulting image (Figure 3) does not give any clue about the solid objects that made up the original still life, the process required an understanding of Kandinsky's analytical method.

Below is an example of NST presented in the TensorFlow tutorial, where Kandinsky's Composition 7 was used as a style reference. The style of an image is described by the means and correlations across the different feature maps in this technique [Gatys et al., 2015].

However, we believe this application comes from misunderstanding about the nature of the original painting. Compared to our initial outcome from this project, it did not make any attempt to understand the artist's intention or process of making abstract art. As can be seen from this example, it requires careful investigation when replicating aspects of creative artistic behavior.

6 DISCUSSION

It has been a great interest for people to produce computer programs that can be used to assist and augment human creativity, especially in the fields of music and visual arts [López de Mántaras, 2016]. Eric Kandel, the Nobel Prize-winning neuroscientist, mentioned in his book, *Reductionism in Art and Brain Science: Bridging the Two Cultures* [Kandel, 2016], that many abstractionists effectively created new rules for visual processing. He argues that abstract art is therefore the key to understanding how art and science together might open entirely new ways of seeing and imagining.

Through the implementation of analytical drawing, we believe we gained a better understanding of the elaborate process of formulating a composition in abstract images and synthesizing them. We hope to continue this work to gain a fuller understanding of the realm of abstract art.

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A THREE STAGES [POLING, 1986]

A SIMPLIFICATION

The first stage required the students: (1) to subordinate the whole complex to one simple overall form, which... must be precisely drawn in; (2) to realize the formal characterization of individual parts of the still life, regarded both in isolation and in relation to the other parts; and (3) to represent the whole construction by means of the most concise possible schema.

B ANALYSIS

The second stage was designated "development of the structural network". Its tasks were: (1) making clear the tensions discovered in the structure, which are to be represented by means of linear forms; (2) emphasizing the principal tensions by means of broader lines or, subsequently, colors; and (3) indicating the structural network by means of starting or focal points (dotted lines).

C TRANSFORMATION

The third stage advances the aspects of the second toward more radical, freer, abstract solutions. It is characterized as: (1) Objects are regarded exclusively in terms of tensions between forces, and the construction limits itself to complexes of lines; (2) Variety of structural possibilities: clear and concealed construction; and (3) Exercises in the utmost simplification of the overall complex and of the individual tensions-concise, exact expression.